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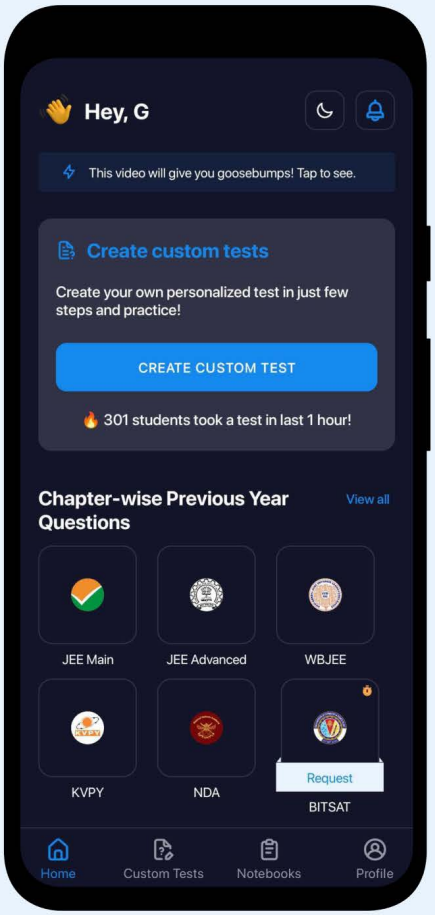

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PROBABILITY

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PROBABILITY

To begin with, you all must have heard of this term though not in academic sense but relating to many other events in your life. Some famous examples are

1. In Betting, chances of winning of India.
2. While playing Lodu, the probability of occuring 6 on the face of a dice.
3. While playing cards, etc.

This chapter is one of those Chapters which will be utilized throughout your life irrespective of academics. Notion is that this is a very difficult chapter which is wrong. This is an easy though TRICKY chapter.

So without beating around the bush we will come directly to the point and will keep the approach as direct as possible.

1. SOME BASIC DEFINITIONS

1.1 Experiment

An experiment is an operation which can result in more than one way.

Random Experiment : An experiment whose outcome can not be predicted with certainty is called a random experiment.

Examples : (i) "Throwing an unbiased die" is a random experiment because when a die is thrown we can not say with certainty which one of the numbers 1, 2, 3, 4, 5 and 6 will come up.

(ii) "Tossing of a fair coin" is a random experiment because when a coin is tossed we can not say with certainty whether either a head or a tail will come up.

(iii) "Drawing a card from a well-shuffled pack of cards" is a random experiment.

Note : (i) A pack of cards consists of 52 cards in 4 suits i.e.,

(a) spades (♠) (b) Clubs (♣)

(c) Hearts (♥) (d) Diamonds (♦).

Each suit consists of 13 cards. Out of these, spades and clubs are black faced cards, while hearts and diamonds are red-faced cards. The aces, kings, queens, Jack (or knave) are called face cards or honour cards.

1.2 Sample Space

The set of all possible results of a random experiment is called the sample space of that experiment and it is generally denoted by S.

Examples : (i) If we toss a coin, there are two possible results, namely a head (H) or a tail (T).

So, the sample space in this experiment is given by

S = {H, T}.

(ii) When two coins are tossed, the sample space

S = {HH, HT, TH, TT}

where HH denotes the head on the first coin and head on the second coin, Similarly HT denotes the head on the first coin and tail on the second coin.

1.3 Event

An event is a subset of the sample space S.

Example :

Let us consider the experiment of tossing a coin. This experiment has two possible outcomes : heads (H) or tails (T).

$\Rightarrow$ sample space = S = {H, T}

We can define one or more events based on this experiment. Let us define the two events A and B as :

A : heads appears B : tails appears

It is easily seen that set A (corresponding to event A) contains outcomes that are **favourable to event A** and set B contains **outcomes favourable to event B.** Recalling that n(A) represents the number of elements in set A, we can observe that

n(A) = number of outcomes favourable to event A

n(B) = number of outcomes favourable to event B

$\Rightarrow$ n(S) = number of possible outcomes

Here in this example, n(A) = 1 ; n(B) = 1 and n(S) = 2.

(ii) When we throw a die then any one of the numbers 1, 2, 3, 4, 5 and 6 will come up. So the sample space S = {1, 2, 3, 4, 5, 6}

1.4 Elementary Event

An event having only a single sample point is called an elementary or simple event.

For example When two coins are tossed,the sample space S = {HH, HT, TH, TT} then the event E_1 = {HH} of getting both the heads is a simple event.

1.5 Mixed Event or Compound Event or Composite Event

An event other than elementary or simple event is called mixed event.

For example (i) When two coins are tossed,the sample space

S = {HH, HT, TH, TT}

Then the event E = {HH, HT, TH} of getting at least one head, is a mixed event.

(ii) When a die is thrown, the sample space S = {1, 2, 3, 4,5, 6}

Let A = {2, 4, 6} = the event of occurrence of an even number

B = {3, 6} = the event of occurrence of a number divisible by 3.

Here A and B are mixed events.

1.6 Equally likely Events

The given events are said to be equally likely if none of them is expected to occur in prefernce to the other.

For example : (i) When an unbiased coin is tossed then occurrence of head or tail are equally likely cases and there is no reason to expect a 'head' or a 'tail' in preference to the other.

(ii) When an unbiased die is thrown, all the six faces 1, 2, 3, 4, 5 and 6 are equally likely to come up. There is no reason to expect 1 or 2 or 3 or 5 or 6 in preference to the other.

1.7 Independent Events :

Two events are said to be independent, if the occurrence of onedoes not depend on the occurrence of the other.

Let $E_1 = \{1, 3, 5\}$ = the event of occurrence of an odd number

and $E_2 = \{2, 4, 6\}$ = the event of occurrence of an even number.

Clearly, the occurrence of odd number does not depend on the occurrence of even number.

So, E_1 and E_2 are independent events.

1.8 Complementary Event :

Let E be an event and S be the sample space for a random experiment then complement of E as denoted by E′ or E^c or $\overline{E}$. Clearly, E′ means E does not occur.

Thus E′ occurs ⇔ E does not occur.

For example When an unbiased die is thrown then the sample space S = {1, 2,3, 4, 5, 6}.

If $E = \{1, 3, 6\}$ then $E' = \{2, 3, 5\}$

1.9 Exhaustive Cases (Events) :

For a random experiment A, set of cases (events) is said to be exhaustive if one of them must necessarily happen everytime the experiment is performed. For example whena die is thrown cases (events) 1,2, 3, 4, 5, 6, form an exhaustive set of cases (events.)

Probability of occurrence of an event :

I. Mathematical or a Priori or Classical Definition of Probability :

Let S be the sample space, then the probability of occurrence of an event E is denoted by P(E) and is defined as

$$P(E) = \frac{n(E)}{n(S)} = \frac{\text{number of elements in E}}{\text{number of elements in S}}$$

$$= \frac{\text{number of cases favourable to event E}}{\text{total number of cases}}$$

Event

(i) When a coin is tossed, sample space S = {H, T}

Let E = the event of occurrence of a head.

Then E = {H}

$\therefore$ P(E) = Probability of occurrence of head $= \frac{n(E)}{n(S)} = \frac{1}{2}$

(ii) When a die is tossed, sample space S = {1, 2, 3, 4, 5, 6}

Let A = the event of occurrence of an odd number = {1, 3, 5}

B = the event of occurrence of a number greater than 4 = {5, 6}.

Then $P(A) = \frac{n(A)}{n(S)} = \frac{3}{6} = \frac{1}{2}$

and $P(B) = \frac{n(B)}{n(S)} = \frac{2}{6} = \frac{1}{3}$

(ii) When one ball is drawn at random from a bag containing 3 black and 4 red balls (balls of the same colour being identical or different), then sample space S = $\{B_1, B_2, B_3, R_1, R_2, R_3, R_4\}$.

$\therefore$ $n(S) = 7$

Here the three black balls may be denoted by B_1, B_2 and B_3 even if they are identical because while finding probability only number of black and red balls are to be taken into account.

Let E = the event of occurrence of a red ball.

Then E = $\{R_1, R_2, R_3, R_4\}$ $\therefore$ n(E) = 4

Now $P(E) = \frac{n(E)}{n(S)} = \frac{4}{7}$

(iii) When two coins are tossed, sample space S = {HH, HT, TH, TT}.

Let E = the event of occurrence of one head and one tail, then

E = {HT, TH}

Now $P(E) = \frac{n(E)}{n(S)} = \frac{2}{4} = \frac{1}{2}$

Theorem 1. : Probability of occurrence of an event is a number lying between 0 and 1.

$$0 \le P(E) \le 1$$

Note : (i) If ϕ is the impossible event,

then $P(\phi) = \frac{n(\phi)}{n(S)} = \frac{0}{n(S)} = 0$

(ii) If S be the sure event, then $P(S) = \frac{n(S)}{n(S)} = 1$.

(iii) $P(E) = 0 \Leftrightarrow E = \phi$

and $P(E) = 1 \Leftrightarrow E = S$

Theorem 2. : If is any event and E´ be the complement of event E, then P(E) + P(E´) = 1.

Odds in Favour and Odds Against an Event :

Let S be the sample space and E be an event. Let E´ denote the complement of event E, then

(i) odds in favour of even E

$$= \frac{n(E)}{n(E')} = \frac{\text{number of cases favourable to event E}}{\text{number of cases against E}}$$

(ii) odds against an event E

$$= \frac{n(E')}{n(E)} = \frac{\text{number of cases against the event E}}{\text{number of cases favourable to event E}}$$

From definitions given above it is clear that,

Note : (i) Odds in favour of event E

$$= \frac{n(E)}{n(E')} = \frac{n(E)/n(S)}{n(E')/n(S)} = \frac{P(E)}{P(E')}$$

and odds against event E $= \frac{n(E')}{n(E)} = \frac{P(E')}{P(E)}$

(ii) If any one of P(E), odds in favour of E and odds against E is given, then other two can be determined.

If odds in favour of an event are a : b then the probability of the occurrence of that event is a/(a + b) and of non-occurrence is b/(a + b).

Ex. (i) If $P(E) = \frac{2}{7}$, then odds in favour of E = $\frac{2}{5}$ and odds against E = $\frac{5}{2}$.

(ii) If odds against E = $\frac{3}{11}$, then odds in favour of E = $\frac{11}{3}$ and $P(E) = \frac{11}{14}$

(iii) If odds in favour of E = $\frac{3}{8}$, then odds against E = $\frac{8}{3}$ and P(E) = $\frac{3}{11}$

Illustration 1

If two dice are thrown simultaneously, represent the sample space and the following events.

(i) **the sum of the numbers coming up is greater than 9.**

(ii) **The sum of the numbers coming up is 8.**

Solution :

Here random experiment is : Throwing of two dice simultaneously.

Let S = the sample space

A = the event that the sum of the numbers coming up is not less than 9.

and B = the event that the sum of the numbers coming up is 8.

Let 1, 2, 3, 4, 5 and 6 denote the occurrence of numbers 1, 2,3, 4, 5, 6 respectively on a die.

Let (m, n) denote the occurrence of m on first die and n on second die, then

S = {(1, 1), (1, 2), (1, 3), (1, 4), (1,5), (1, 6), (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6), (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6), (5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6), (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)}

A = {(6, 3), (5, 4), (4, 5), (3, 6), (6, 4), (5, 5), (4, 6), (6, 5), (5, 6,), (6, 6)}

and B = {(6, 2), (5, 3), (4, 4), (3, 5), (2, 6)}

Illustration 2

If three coins are tossed, represent the sample space and the event of getting two heads and one tail and also find the number of elements in them.

Solution :

Here random experiment is : tossing of three coins.

Let S = the sample space

and E = the event of occurrence of two heads and one tail.

Let H denote the occurrence of head and T the occurrence of tail when one coin is tossed.

Then S = {H, T} {H, T} {H, T}

= {(H, H, H), (H, H, T), (H, T,H), (T, H, H), (H, T, T), (T, H, T), (T, T, H), (T, T, T)}

and E = {(H, H, T), (H, T, H), (T, H, H)}

Also n(S) = 8 and n(E) = 3.

Illustration 3

A coin is tossed successively three times. Find the probability of getting exactly one head or two heads.

Solution :

Let S be the sample space and E be the event of getting exactly one head or exactly two heads, then

S = {HHH, HHT, HTH, THH, TTH, THT, HTT, TTT}.

and E = {HHT, HTH, THH, HTT, THT, TTH}

∴ n(E) = 6 and n(S) = 8.

Now required probability, $P(E) = \frac{n(E)}{n(S)} = \frac{6}{8} = \frac{3}{4}$

Illustration 4

Two dice are thrown simultaneously. What is the probability of obtaining a total score of 7 ?

Solution :

Here random experiment is : throwing of two dice at a time

Let S be the sample space and E be the event of obtaining a total of 7.

Then S = {1, 2, 3, 4, 5, 6} {1, 2, 3,4, 5, 6}

$\therefore$ n(S) = 6 6 = 36

Also E = {(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)}

$\therefore$ n(E) = 6

Now **required probability,** $P(E) = \frac{n(E)}{n(S)} = \frac{6}{36} = \frac{1}{6}$

Note here that (1, 6) and (6, 1) are 2 different cases

Illustration 5

If a leap year is selected at random, what is the chance that it will contain 53 Tuesdays ?

Solution :

A leap year has 366 days i.e. 52 complete weeks and two days more. These two days will be the two consecutive days of a week. A leap year will have 53 Tuesdays if out of the two consecutive days of a week selected at random one is a Tuesday.

Here random experiment is : Selection of two consecutive days of a week.

Let S be the sample space and E be the event that out of the two consecutive days of a week one is a Tuesday, then

S = {(Monday, Tuesday), (Tuesday, Wednesday),
(Wednesday, Thursday), (Thursday, Friday),
(Friday, Saturday), (Saturday, Sunday), (Sunday, Monday)}

$\therefore$ n(S) = 7

and E = {(Monday, Tuesday), (Tuesday, Wednesday)}

$\therefore$ n(E) = 2

Now, **required probability,** $P(E) = \frac{n(E)}{n(S)} = \frac{2}{7}$

Illustration 6

What are the odds in favour of throwing at least 8 in a single throw with two dice ?

Solution :

Here random experiment is : throwing of two dice.

Let S be the sample space and E the event of occurrence of a total of at least 8, then

S = {1, 2, 3, 4, 5, 6} {1, 2, 3, 4, 5, 6}

$\therefore$ n(S) = 6 6 = 36

Throwing at least 8 with two dice means throwing 8 or 9 or 10 or 11 or 12.

$\therefore$ E = {(6, 2), (5, 3), (4, 4), (3, 5), (2, 6), (6, 3), (5, 4), (4, 5), (3, 6), (6, 4), (5, 5), (4, 6), (6, 5), (5, 6), (6, 6)}

$\therefore\ n(E) = 15$

$\therefore\ n(E') = 36 - 15 = 21$

$\therefore$ **odds in favour of** $E = \dfrac{n(E)}{n(E')} = \dfrac{15}{21} = \dfrac{5}{7} = 5 : 7$

Illustration 7

Two dice are thrown simultaneously. What is the probability obtaining a total score less than 11 ?

Solution :

Let S be the sample space and E be the event of obtaining a total less than 11.

Then $S = \{1, 2, 3, 4, 5, 6\} \quad \{1, 2, 3, 4, 5, 6\}$

$\therefore \quad n(S) = 6 \quad 6$

$= 36$

then E′ be the event of obtaining a total score greater than or equal to 11.

Also $E' = \{(5, 6), (6, 5), (6, 6)\}$

$\therefore \quad n(E') = 3$

The Probability of obtaining a total score greater than or equal to 11,

$$P(E') = \frac{n(E')}{n(S)} = \frac{3}{36} = \frac{1}{12}$$

$$\therefore \quad P(E) = 1 - P(E')$$

$$= 1 - \frac{1}{12}$$

$$= \frac{11}{12}$$

Hence required probability is $\dfrac{11}{12}$.

Illustration 8

Balls are drawn one-by-one without replacement from a box containing 2 black, 4 white and 3 red balls till all the balls are drawn. Find the probability that the balls drawn are in the order 2 black, 4 white and 3 red. [IIT 1978]

Solution :

To draw 2 black,4 white and 3 red balls in order is same as arranging two black balls at first 2 places, 4 white at next 4 places, (3rd or 6th place) and 3 red at still next 3 places (7th to 9th

place), i.e., $B_1B_2\ W_1W_2W_3W_4\ R_2R_2R_3$, which can be done in 2 ! 4 ! 3 ! ways. And total ways of arranging all 2 + 4 + 3 = 9 balls is 9 !

$\therefore$ Required probability $= \dfrac{2! \times 4! \times 3!}{9!} = \dfrac{1}{1260}$

Illustration 9

Six boys and six girls sit in a row randomly. Find the probability that

(i) **the six girls sit together**

(ii) **the boys and girls sit alternately.** [IIT 1979]

Solution :

(i) 6 boys and 6 girls sit in a row randomly.

Total ways of their seating = 12 !

No. of ways in which all the 6 girls sit together = 6 ! 7 ! (considering all 6 girls as one person)

$\therefore$ Probability of all girls sitting together $= \dfrac{6\,! \times 7\,!}{12\,!}$

$$= \frac{720}{12 \times 11 \times 10 \times 9 \times 8} = \frac{1}{132}$$

(ii) Starting with boy, boys can sit in 6 ! ways leaving one place between every two boys and one at last.

B _ B _ B _ B _ B _

These left over places can be occupied by girls in 6 ! ways.

$\therefore$ If we start with boys, no. of ways of seating boys and girls alternately = 6 ! 6 !

In the similar manner, if we start with girl, no. of ways of seating boys and girls alternately

= 6 ! 6 !

G _ G _ G _ G _ G _ G _

Thus total ways of alternate seating arrangements

= 6 ! 6 ! + 6 ! 6 !

= 2 6 ! 6 !

$\therefore$ Probability of making alternate seating arrangement for 6 boys and 6 girls

$$= \frac{2 \times 6\,! \times 6\,!}{12\,!}$$

$$= \frac{2 \times 720}{12 \times 11 \times 10 \times 9 \times 8 \times 7} = \frac{1}{462}$$

Illustration 10

Out of 3n consecutive integers, three are selected at random. Find the probability that their sum is divisible by 3.

Solution :

Let the sequence of 3n consecutive integers begins with the integer m. Then the 3n consecutive integers are

$$m,\ m + 1,\ m + 2,\ ....,\ m + (3n - 1).$$

Out of these integers, 3 integers can be chosen in ${}^{3n}C_3$ ways.

Let us divide these 3n consecutive integers into three groups G_1, G_2 and G_3 as follows :

G_1 : m, m + 3, m + 6, ..., m + (3n – 3)

G_2 : m + 1, m + 4, m + 7, ..., m + (3n – 2)

G_3 : m + 2, m + 5, m + 8, ..., m + (3n – 1)

The sum of 3 integers chosen from the given 3n integers will be divisible by 3 if either all the three integers are chosen from the same group or one integer is chosen from each group. The number of ways that the three integers are from the same group is $({}^nC_3 + {}^nC_3 + {}^nC_3)$ and the number of ways that the integers are from different groups is $({}^nC_1 \quad {}^nC_1 \quad {}^nC_1)$.

So, the number of ways in which the sumof three integer is divisible by 3 is $({}^nC_3 + {}^nC_3 + {}^nC_3) + ({}^nC_1 \quad {}^nC_1 \quad {}^nC_1)$.

Hence, required probability $= \dfrac{3\ .\ {}^nC_3 + ({}^nC_1)^3}{{}^{3n}C_3} = \dfrac{3n^2 - 3n + 2}{(3n-1)(3n-2)}$

Illustration 11

Two numbers a and b are chosen at random from the set (1, 2, 3, ...,3n). Find the probability that (i) $a^2 - b^2$ is divisible by 3.(ii) $a^3 + b^3$ is divisible by 3.

Solution :

The number of ways of choosing two numbers from the given set is ${}^{3n}C_2$.

Let us divide given 3n numbers into three groups G_1, G_2 and G_3 as follows :

G_1 : 3, 6, 9, ..., 3n

G_2 : 1, 4, 7, 10, ..., 3n – 1

G_3 : 2, 5, 8, 11, ..., 3n – 2

(i) Now, $a^2 - b^2 = (a - b)(a + b)$

Therefore, $a^2 - b^2$ will be divisible by 3 if either a and b are chosen from the same group or one of them is chosen from G_2 and the other from G_3. Therefore, the favourable number of ways is $({}^nC_2 + {}^nC_2 + {}^nC_2) + {}^nC_1 \quad {}^nC_1 = 3\ {}^nC_2 + n^2$

Hence, the required probability $= \dfrac{3\ .\ {}^nC_2 + n^2}{{}^{3n}C_2} = \dfrac{5n - 3}{3\,(3n-1)}$

(ii) We have, $a^3 + b^3 = (a + b)^3 - 3ab\ (a + b)$

Therefore, $a^3 + b^3$ will be divisible by 3, if a + b is divisible by 3. Now, a + b will be divisible by 3 in the following cases :

I. Both the numbers belong to the first group.

II. One of the two numbers belongs to second groupand one of them belongs to the third group.

Therefore, favourable number of ways = ${}^nC_2 + {}^nC_1 \quad {}^nC_1$

Hence, the required probability $= \dfrac{{}^nC_2 + {}^nC_1 \times {}^nC_1}{{}^{3n}C_2} = \dfrac{1}{3}$

Illustration 13

Six dice are thrown simultaneously. Find the probability that

(i) **all of them show the same face.**

(ii) **all of them show different faces.**

(iii) **exactly three of them show the same face and remaining three show different faces.**

(iv) **atleast four of them show the same face.**

Solution :

The total number of elementary events associated to the random experiment of throwing six dice is $6 \quad 6 \quad 6 \quad 6 \quad 6 \quad 6 = 6^6$.

(i) All dice show the same face means we are getting same number on all six dice. The number of ways for which is ${}^6C_1$.

Hence, required probability $= \dfrac{{}^6C_1}{6^6} = \dfrac{1}{6^5}$.

(ii) The total number of ways in which all dice show different faces is same as the numberof arrangements of 6 numbers 1, 2, 3, 4, 5, 6 by taking all at a time.

So, favourable number of elementary events = 6 !

Hence, required probability $= \dfrac{6\,!}{6^6}$.

(iii) Select a number which occurs on three dice out of six numbers 1, 2, 3, 4, 5, 6. This can be done in ${}^6C_1$ ways. Now select three numbers out of the remaining 5 numbers. This can be done in ${}^5C_3$ ways. Now we have 6 numbers like 1, 2, 3, 4, 4, 4; 2, 3, 6, 1, 1, 1 etc. These digits can be arranged in $\dfrac{6\,!}{3\,!}$ ways. So, the number of ways in which three dice show the same face and the remaining three show distinct faces is

$${}^6C_1 \quad {}^5C_3 \quad \frac{6\,!}{3\,!}$$

Hence, required probability $= \dfrac{{}^6C_1 \times {}^5C_3 \times \dfrac{6!}{3!}}{6^6}$

(iv) Atleast four of them show the same face means either four dice show the same face and remaining two show distinct faces or five dice show the same face and remaining one shows a different face or all the six show same face. So, required probability

$$= \frac{{}^6C_1 \times {}^5C_2 \times \frac{6!}{4!} + {}^6C_1 \times {}^5C_1 \times \frac{6!}{5!} + {}^6C_1}{6^6}$$

Illustration 14

If the integers m **and** n **are choosen at random between 1 and 100, then find the probability that a number of the form $7^m + 7^n$ is divisible by 5.**

Solution :

We know that

$7^1 = 7,\ 7^2 = 49,\ 7^3 = 343,\ 7^4 = 2401,\ 7^5 = 16807$

$\therefore$ 7^k (where $k \in Z$), results in a number whose unit's digit is 7 or 9 or 3 or 1.

Now, $7^m + 7^n$ will be divisible by 5 if unit's place digit of resulting number is 5 or 10 clearly it can never be 5.

But it can be 0 if we consider values of m and n such that the sum of unit's place digits become 0. And this can be done by choosing

m = 1, 5, 9, ... 97

and corresondingly

n = 3, 7, 11, ... 99

(25 options each) [7 + 3 = 10]

or m = 2, 6, 10, ... 98

n = 4, 8, 12, ... 100

(25 options each) [9 + 1 = 10]

I case. Thus m can be chosen in 25 ways and n can be chosen in 25 ways.

II. case. or m can be chosen in 25 ways and n can be chosen in 25 ways.

$\therefore$ Total no.of selections of m, n so that

$7^m + 7^n$ is divisible by 5 = (25 25 + 25 25) 2

because we can interchange the values of m and n.

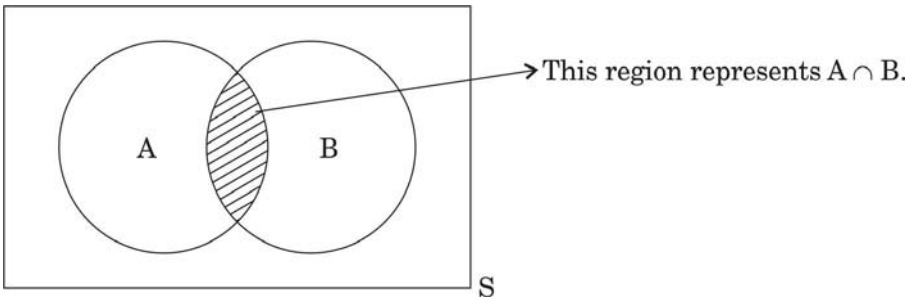
And using the same lgoic, total possible solutions of m and n out of 100 = 100 100

$\therefore$ Required probability $= \dfrac{2\,(2 \times 25 \times 25)}{100 \times 100} = \dfrac{1}{4}$

USE OF VEIN DIAGRAMS

1. $(A \cap B)$

Interpretation : A occurs as well as B occurs or Both A and B occur simultaneously.



Example :

If sample space is, $S \equiv \{1, 2, 3, 4, 5, 6\}$

$A \equiv \{1, 3, 5\}$

$B \equiv \{3, 6\}$

$$P(A \cap B) = \frac{n(A \cap B)}{n(S)}$$

So from example, $n(A \cap B) = \{3\} = 1$ because only 3 is common in both sets

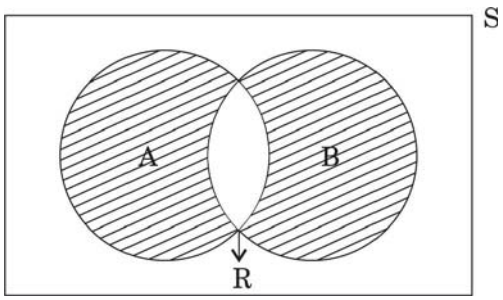
$n(S) = 6$

$\therefore \quad P(A \cap B) = \frac{1}{6}$

2. $(A \cup B)$

Interpretation : Atleast one of A and B occur or either A or B or both occur.

Note : Here case of both can also be there in term of set, it is UNION of A & B.



Shaded Area represents the region required.

Formula : $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

why $- P(A \cap B)$?

This is because we have counted the region R (which represent $A \cap B$) twice in P(A) and P(B). P(A) includes the area 'R' (i.e. $A \cap B$) and P(B) also includes the same. So we have to subtract it once. How to calculate it then ?

Let us take the sample example as taken earlier

$$S \equiv \{1, 2, 3, 4, 5, 6\}$$
$$A \equiv \{1, 3, 5\}$$
$$B \equiv \{3, 6\}$$

$$P(A) = \frac{3}{6} = \frac{1}{2} \quad P(B) = \frac{2}{6} = \frac{1}{3}$$

$$P(A \cap B) = \frac{n(A \cap B)}{n(S)} = \frac{1}{6}$$

$$\therefore \quad P(A \cup B) = \frac{1}{2} + \frac{1}{3} - \frac{1}{6}$$

$$= \frac{1}{2} + \frac{1}{6}$$

$$= \frac{4}{6} = \frac{2}{3}$$

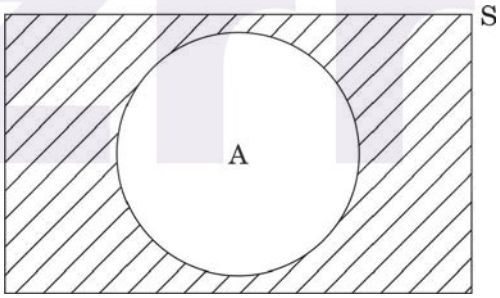
(a) **<u>Complementary Events</u> :**

i.e. non-occurence of A (the shaded area in the diagram)

So if P(A) is the probability of occurence of A then

$P(\overline{A}) = 1 - P(A)$

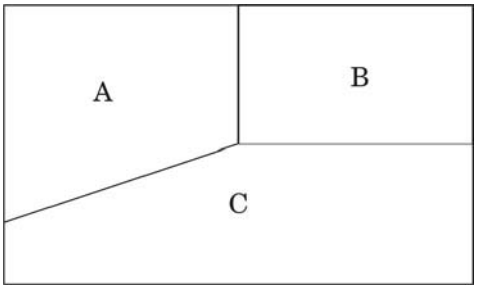
as the entire space (in diagram entire rectangle has probability of occurence as 1.



(b) **<u>Exhaustive Events</u>**

A set of events is said to be exhaustive of atleast one of them happens when experiment takes place.

OR, the sample space is completely filled up by events (as shown in diagram)



Condition :

$$P(A \cup B \cup C) = 1$$

(c) **<u>Mutually Exclusive Events</u> :**

A and B are two mutually exclusive events, if happening of A rules out the happening of B or vice versa i.e. they cannot occur simultaneously in terms of set

disjoint of A &B is null set i.e., $A \cap B = \phi$

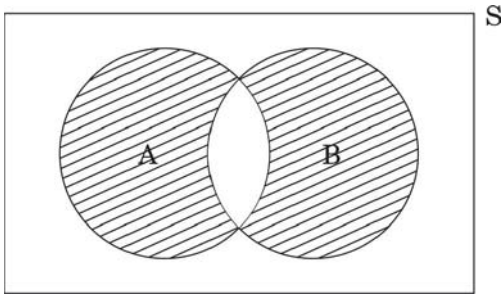
Condition :

$P(A \cup B) = P(A) + P(B)$

in general

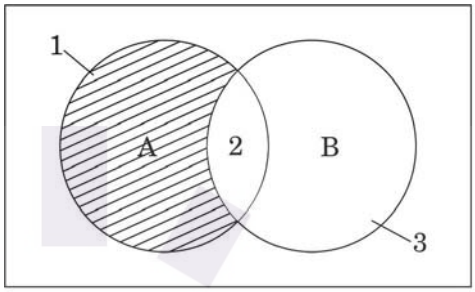
$P(A_1 \cup A_2 .... A_n) = P(A_1) + P(A_2) + ... P(A_n)$

where $A_1, A_2, A_3 .... A_n$ are n mutually exclusively events from diagram the area where A & B do not occur simultaneously is the answer.

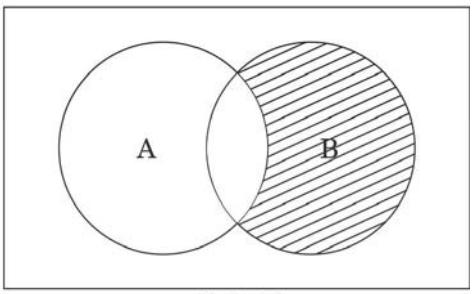


3. Only A

Interpretation : This means that only one of the events occur at a time.



Only A



Only B

Let us take the case of only A to understand it

represents : $P(A \cap \overline{B})$

Logic : We want the region of only A and it we see the vein diagram there is a region $A \cap B$, i.e. some part of B is also there hence $P(A \cap \overline{B})$, area of A not containing (i.e. $\overline{B}$) B.

formula : $P(A \cap \overline{B}) = P(A) - P(A \cap B)$

$= \{\text{region } (1 + 2)\} - \{\text{region } 2\}$

4. Exactly one of A and B

Interpretation : It means that either only A occur or only B occurs, but no occurence of event containing both A & B.

Shaded region represents it

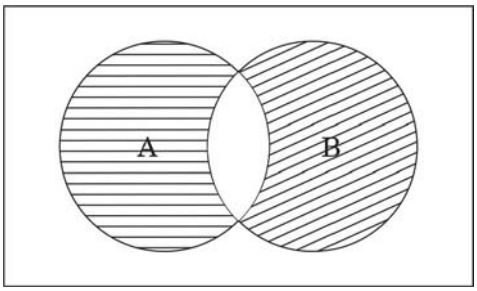
Note : No $(A \cap B)$ region included.

formula $= P(\text{only } A) + P(\text{only } B)$

$= P(A \cap \overline{B}) + P(\overline{A} \cap B)$

$= \{P(A) - P(A \cap B)\} + \{P(B) - P(A \cap B)\}$

$= P(A) + P(B) - 2\,P(A \cap B)$



as $(P(A) + P(B) = (P(A \cup B) + P(A \cap B)$

$= P(A \cup B) - P(A \cap B)$

Also you can just simply derive the formula from vein diagram.

shaded region $= P(A \cup B) - P(A \cap B)$

example, extending the earlier one

we know $P(A \cup B) = \frac{2}{3}$

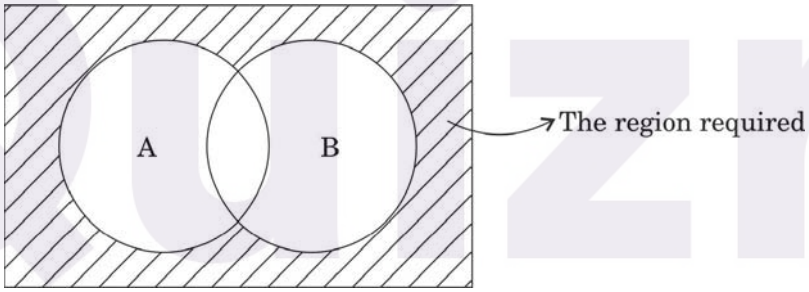
and $P(A \cap B) = \frac{1}{6}$

$\therefore$ Required Answer $= \frac{2}{3} - \frac{1}{6}$

$= \frac{3}{6} = \frac{1}{2}$

5. None of A and B

Interpretation : It means that neither A nor B occurs.



formula $= P(\overline{A} \cap \overline{B})$

$= P(\overline{A} \cup \overline{B})$ (you can just see from the above diagram)

$= 1 - P(A \cup B)$

Note : Now since we have covered so many types let us discuss another formula for $P(A \cup B)$.

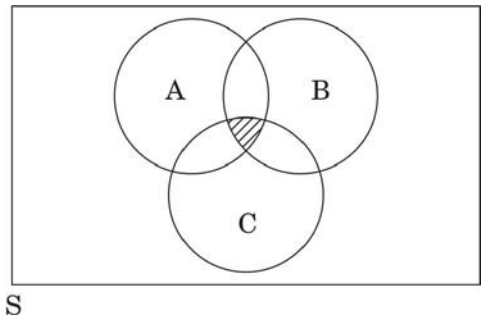
$P(A \cup B) = P(\text{only } A) + P(\text{only } B) + P(\text{Both } A \text{ and } B)$

$= P(A \cap \overline{B}) + P(\overline{A} \cap B) + P(A \cap B)$

1. $\boxed{P(A \cap B \cap C)}$

Interpretation : Occurrence of all the events A, B and C simultaneously.

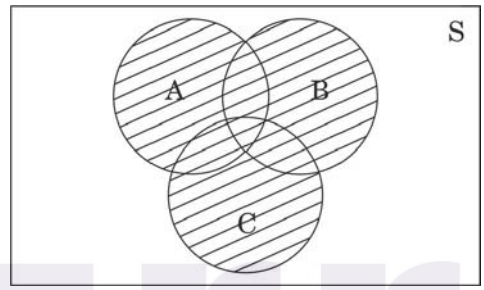
$$P(A \cap B \cap C) = \frac{n(A \cap B \cap C)}{n(S)}$$



2. $\boxed{P(A \cup B \cup C)}$

Interpretation : either A or B or C occurs or all of them can occur

$$P(A \cup B \cup C) = P(A) + P(B) + P(C)$$
$$- P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$



Why add region 1 ?

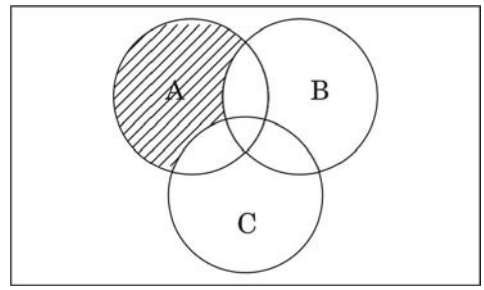
We have subtracted the region 1 3 times in $P(A \cap B)$, $P(B \cap C)$ and $P(C \cap A)$, hence we need to add it to that $P(A \cap B \cap C)$, i.e. event where all can occur can be counted.

3. $\boxed{\text{Only A}}$

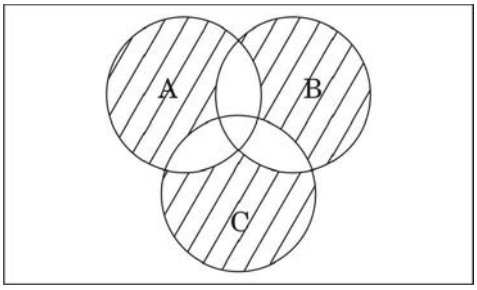
As done in case of 2 event P(only A) = $P(A \cap \overline{B} \cap \overline{C})$ can say

$$P(\text{only A}) = P(A) - P(A \cap B) - P(A \cap C) + P(A \cap B \cap C)$$

Note : Again we have added the part $P(A \cap B \cap C)$ logic is same as explained in previous part. Think it for yourself.



4. Exactly 1 event occurs :



this is equal to

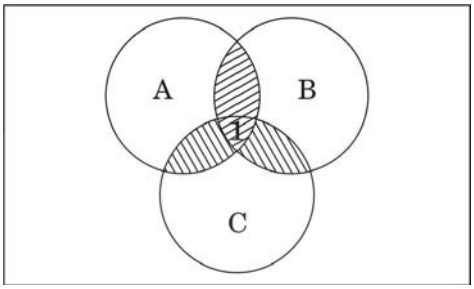
P(only A) + P(only B) + P(only C)

$= P(A \cap \overline{B} \cap \overline{C}) + P(\overline{A} \cap B \cap \overline{C}) + P(\overline{A} \cap \overline{B} \cap C)$

put the values using the above result to get

$= P(A) + P(B) + P(C) - 2P(A \cap B) - 2\,P(B \cap C)$
$\quad - 2\,P(C \cap A) + 3\,P(A \cap B \cap C)$

5. Exactly 2 events occur :



Interpretation :

either A and B occur or B and C occur or C and A occur.

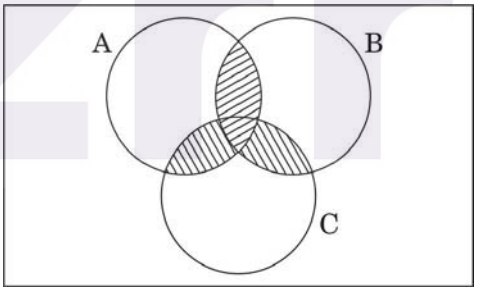
Note : Region 1 is not included here.

So from vein diagram, deriving the formula

$P(\text{exactly } 2) = P(A \cap B) + P(B \cap C) + P(C \cap A)$
$\qquad - 3P\,(A \cap B \cap C)$

again notice the subtraction of $P(A \cap B \cap C)$, as all terms contain it, so have to subtract it.

6. Atleast 2 events occur :



Interpretation : This means that minimum of 2 events should occur. So 3 events occuring simultaneously is perfectly fine in this case

$P\,(\text{atleast } 2) = P(A \cap B) + P(B \cap C) + P(C \cap A)$
$\qquad - 2P(A \cap B \cap C).$

Illustration 15

Two cards are drawn at random from a pack of cards. Find the probability that both the cards are of red colour or they are queen.

Solution :

Here random experiment is :

Drawing of two cards from a pack of 52 cards.

Let S = the sample space.

A = the eventhat the two cards drawn are red.

B = the event that the two cards drawn are Queen.

Total number of cards = 52, number of red cards = 26, number of queen = 4

Now, $n(S) = {}^{52}C_2$, $n(A) = {}^{26}C_2$, $n(B) = {}^{4}C_2$

$A \cup B$ = the event that the two cards drawn are both red or both queen.

$A \cap B$ = the event that the two cards drawn are queen of red colour.

Required probability,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{{}^{26}C_2}{{}^{52}C_2} + \frac{{}^{4}C_2}{{}^{52}C_2} - \frac{{}^{2}C_2}{{}^{52}C_2}$$

$$= \frac{26 \times 25}{52 \times 51} + \frac{4 \times 3}{52 \times 51} - \frac{1 \times 2}{52 \times 51} = \frac{285}{2652}$$

Illustration 16

20 tickets are numbered from 1 to 20. One ticket is drawn at random out of these tickets. What is the probability that the ticket number is a multiple of 2 or 5 ?

Solution :

Here random experiment is : Drawing of one ticket numbered from 1 to 20.

Let S = the sample space.

A = the event that the ticket drawn contains a number which is a multiple of 2.

B = the event that the ticket drawn contains a number which is a multiple of 5.

Then $A \cup B$ = the event that the ticket drawn contains a number which is a multiple of 2 or 5.

and $A \cap B$ = the event that the ticket drawn contains the number which is a multiple of 2 and 5 i.e. a multiple of 10.

Now,

$$S = \{1, 2, 3, 4, ..., 20\} \therefore n(S) = 20$$

$$A = \{2, 4, 6, 8, 10, 12, 14, 16, 18, 20\} \therefore n(A) = 10$$

$$B = (5, 10, 15, 20\} \therefore n(B) = 4$$

$$A \cap B = \{10, 20\} \therefore n(A \cap B) = 2$$

Now,

$$P(A) = \frac{n(A)}{n(S)} = \frac{10}{20}, \; P(B) = \frac{n(B)}{n(S)} = \frac{4}{20}$$

and

$$P(A \cap B) = \frac{n(A \cap B)}{n(S)} = \frac{2}{20}$$

$\therefore$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{10}{20} + \frac{4}{20} - \frac{2}{20} = \frac{12}{20} = \frac{3}{5}$$

Second method : $A \cup B = \{2, 4, 5, 6, 8, 10, 12, 14, 15, 16, 18, 20\}$

$$\therefore\ n(A \cup B) = 12 \ \therefore\ P(A \cup B) = \frac{n(A \cup B)}{n(S)} = \frac{12}{20} = \frac{3}{5}$$

Illustration 17

Two dice are tossed once. Find the probability of getting an even number on first die, or a total of 8.

Solution :

Here random experiment is : Tossing of two dice

Let S be the sample space. Then, n(S) = 36.

Let A = the event of getting an even number on first die

and, B = the event of getting a total of 8.

Then, $A \cap B$ = the event of getting an even number on first die and a total of 8.

Now, $A = \{(2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6), (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\}$;

and $B = \{(2, 6), (6, 2), (3, 5), (5, 3), (4, 4)\}$

$\therefore$ $A \cap B = \{(2, 6), (6, 2), (4, 4)\}$.

$\therefore$ $n(A) = 18$, $n(B) = 5$ and $n(A \cap B) = 3$.

Now, $P(A) = \frac{n(A)}{n(S)} = \frac{18}{36} = \frac{1}{2}$; $P(B) = \frac{n(B)}{n(S)} = \frac{5}{36}$

and $P(A \cap B) = \frac{n(A \cap B)}{n(S)} = \frac{3}{36} = \frac{1}{12}$

Now **required probability** $P(A \cup B) = P(A + P(B) - P(A \cap B) = \left(\frac{1}{2} + \frac{5}{36} - \frac{1}{12}\right) = \frac{20}{36} = \frac{5}{9}$.

Illustration 18

The probability of a horse A winning a race is $\frac{1}{3}$ and that of a horse B winning the same race is $\frac{1}{5}$, find the probability that (i) either of them will win, (ii) none of them will win.

Solution :

Let E_1 = the event that horse A wins the race

and E_2 = the event that horse B wins the race

Then $E_1 \cup E_2$ = the event that either of horses A and B will win the race

$(E_1 \cup E_2)'$ = the event that none of the horses A and B will win the race.

According to question $P(E_1) = \frac{1}{3}$, $P(E_2) = \frac{1}{5}$

Since both the horses A and B take part in the same race, therefore, both of them cannot win at a time.

Hence E_1 and E_2 are mutually exclusive events

$$\therefore \quad P(E_1 \cup E_2) = P(E_1) + P(E_2) = \frac{1}{3} + \frac{1}{5} = \frac{8}{15}$$

and $P(E_1 \cup E_2)' = 1 - P(E_1 \cup E_2) = 1 - \frac{8}{15} = \frac{7}{15}$

Illustration 19

A and B are two candidates seeking admission in I.I.T. The probability that A is selected is 0.5 and the probability that both A and B are selected is at most 0.3. Is it possible that the probability of B getting selected is 0.9. [I.I.T. 1982]

Solution :

Let E_1 = the event that A is selected

E_2 = the event that B is selected

According to question,

$P(E_1) = 0.5$, $P(E_1 \cap E_2) \le 0.3$

By additiontheorem of probability,

$$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$$
$$\ge 0.5 + P(E_2) - 0.3 \quad [\because \text{ Max. Value of } P(E_1 \cap E_2), = 0.3]$$
$$= 0.2 + P(E_2)$$

$\therefore \quad P(E_2) \le P(E_1 \cup E_2) - 0.2 \le 1 - 0.2 = 0.8 \qquad [\because P(E_1 \cup E_2) \le 1]$

Hence $P(E_2) \ne 0.9$

Illustration 20

An urn contains 5 white, 4 block and 3 yellow balls. Four balls are drawn at random. Find the probability that all of them are of the same colour.

Solution :

Here random experiment is : Drawing of 4 balls from an urn containing 5 white, 5 black and 3 yellow balls.

Let S = the sample space

A = the event of drawing 4 white balls

B = the event of drawing 4 black balls

Then $A \cup B$ = the event of drawing 4 balls of the same colour

Total number of balls = 12

$$\therefore \qquad n(S) = {}^{12}C_4 = \frac{12 \times 11 \times 10 \times 9}{\lfloor 4} = 495$$

$$n(A) = {}^{5}C_4 = \frac{\lfloor 5}{\lfloor 4 \lfloor 1} = 5, \; n(B) = {}^{4}C_4 = 1$$

Now $P(A) = \frac{n(A)}{n(S)} = \frac{5}{495}$ and $P(B) = \frac{n(B)}{n(S)} = \frac{1}{495}$

Since events A and B are mutually exclusive

$$\therefore \qquad P(A \cup B) = P(A) + P(B) = \frac{5}{495} + \frac{1}{495} = \frac{6}{495} = \frac{2}{165}$$

Illustration 21

For a post three persons A, B and C appear in the interview. The probability of A being selected is twice that of B and the probability of B being selected is thrice that of C, what are the individual probabilities of A, B, C being selected ?

Solution :

Let E_1, E_2, E_3 be the events of selection of A, B and C respectively.

Let $P(E_3) = x$

Then according toquestion, $P(E_2) = 3P(E_3) = 3x$

and $P(E_1) = 2P(E_2) = 6x$

Since there are only three candidates A, B and C, therefore, one must be selected and exactly one will be selected.

$\therefore \quad P(E_1 \cup E_2 \cup E_3) = 1$

and E_1, E_2, E_3 are mutually exclusive.

Now $1 = P(E_1 \cup E_2 \cup E_3) = P(E_1) + P(E_2) + P(E_3)$

or $\quad 1 = 6x + 3x + x = 10x$

$$\therefore \quad x = \frac{1}{10} \quad \therefore P(E_3) = x = \frac{1}{10} \; . \; P(E_1) = 6x = \frac{6}{10} = \frac{3}{5}, \; P(E_2) = 3x = \frac{3}{10}$$

Illustration 23

The odds in favour of standing first of three students appearing at an examination are 1 : 2, 2 : 5 and 1 : 7 respectively. What is the probability that either of them will stand first.

Solution :

Let the three students be P, Q and R.

Let A, B, C denote the events of standing first of the three students P, Q, R respectively.

Given, odds in favour of A = 1 : 2

odds in favour of B = 2 : 5

and odds in favour of C = 1 : 7

$$\therefore \quad P(A) = \frac{1}{1+2} = \frac{1}{3},\ P(B) = \frac{2}{2+5} = \frac{2}{7},\ P(C) = \frac{1}{1+7} = \frac{1}{8}$$

Since events A, B, C are mutually exclusive

$$\therefore \quad P(A \cup B \cup C) = P(A) + P(B) + P(C)$$

$$= \frac{1}{3} + \frac{2}{7} + \frac{1}{8} = \frac{56+48+21}{168} = \frac{125}{168}$$

Illustration 24

Two cards are drawn from a pack of 52 cards. What is the probability that either both are red or both are kings ?

Solution :

Out of 52 cards, two cards can be drawn in ${}^{52}C_2$ ways.

So, exhaustive number of cases = ${}^{52}C_2$.

Consider the following events :

A = Two cards drawn are red cards; B = Two cards drawn are kings.

$$\text{Required probability} = P(A \cup B)$$

$$= P(A) + P(B) - P(A \cap B) \quad \text{...(i)}$$

Now we shall find P(A), P(B) and P(A $\cap$ B).

There are 26 red cards,out of which 2 red cards can be drawn in ${}^{26}C_2$ ways.

$$\therefore \quad P(A) = \frac{{}^{26}C_2}{{}^{52}C_2}$$

Since there are 4 kings, out of which 2 kings can be drawn in ${}^{4}C_2$ ways.

$$\therefore \quad P(B) = \frac{{}^{4}C_2}{{}^{52}C_2}$$

There are 2 cards which are both red and kings, therefore

$P(A \cap B)$ = Probability of getting 2 cards which are both red and kings.

$$= \text{Probability of getting 2 red kings} = \frac{{}^4C_2}{{}^{52}C_2}$$

From (i)

$$\text{Required probability} = P(A) + P(B) - P(A \cap B)$$

$$= \frac{{}^{26}C_2}{{}^{52}C_2} + \frac{{}^4C_2}{{}^{52}C_2} - \frac{{}^2C_2}{{}^{52}C_2}$$

$$= \frac{325}{1326} + \frac{1}{221} - \frac{1}{1326} = \frac{55}{221}$$

Illustration 25

A candidate has to appear in an examination in three subjects : English, Mathematics and Physics. His entrances of passing in these subjects are 0.5, 0.7 and 0.9 respectively. Find the probability that :

(a) **he passes in at least one of the subjects.**

(b) **he passes exactly in two subjects.**

Solution :

A : he passes in English

B : he passes in Mathematics

C : he passes in Physics

(a) P(he passes in at least one subject) = $P(A \cup B \cup C)$

To calculate $P(A \cup B \cup C)$, use :

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$$

$$= 0.5 + 0.7 + 0.9 - (0.35 + 0.63 + 0.45) + 0.315 = 0.985$$

<u>Alternatively,</u>

it is easy to calculate $P(A \cup B \cup C)$ by :

$$P(A \cup B \cup C) = 1 - P(\overline{A} \cap \overline{A} \cap \overline{C}) = 1 - P(\overline{A}) \quad P(\overline{B}) \quad P(\overline{C})$$

$$= 1 - (1 - 0.5)(1 - 0.7)(1 - 0.9) = 0.985$$

(b) P(He passes exactly in two subjects) = $P(A \cap B \cap \overline{C}) + P(A \cap \overline{B} \cap C) + P(\overline{A} \cap B \cap C)$

$$= 0.5 \quad 0.7\,(1 - 0.9) + 0.5 \quad (1 - 0.7) \quad 0.9 + (1 - 0.5) \quad 0.7 \quad 0.9$$

$$= 0.485$$

INDEPENDENT EVENTS

Two or more events are said to be independent if the probability of occurence of any of them does not change by the occurence of other.

For example :

If two coins are tossed, then the occurence of either heads or tail on the coins is independent. i.e. there can be anything on the either of coins.

But, now consider the scenario.

A bag contains 5 white and 5 black balls.

Now we define 2 events A and B such that

A : first ball drawn is black without replacement.

B : second ball drawn is again black.

here A & B are not independent as the event B depends on the out come of event A. (for example now in case of event B, only 4 black balls are available to choose from)

MUTUALLY EXCLUSIVE AND INDEPENDENT EVENTS

Let us do the definition of mutual exclusive event once more to understand the difference.

Two events are mutually exclusive events if they cannot occur simultaneously.

Students generally confuse the word simultaneously with occurance term in inependent events. They are irrelevant.

Mutual exclusion is the property of events occuring at same stage i.e. at same point of time. Like occurenceof head and tail on a single toss of a coin is an example of mutual exclusion as both cannot occur simultaneously.

Whereas,if we say we toss the coin twice. Head occurs on the first time whereas tail on the second. We call these events independent and not mutual exclusive events.

These is no point of saying that they cannot occur simultaneously when already they have been defined separately in different stages.

RESULTS ON INDEPENDENT EVENTS

1. Two events A and B are independent if and only if

$$P(A \cap B) = P(A)\,P(B)$$

this result can be extended to any number of independent events

i.e., $$P(A \cap B \cap C) = P(A)\,P(B)\,P(C)$$

$$P(A_1 \cap A_2 \cap \ldots A_n) = P(A_1)\,P(A_2)\ldots P(A_n)$$

where $A_1\ A_2 \ldots A_n$ are mutually exclusive events.

2. $P(A_1 \cup A_2 \cup \ldots A_n) = 1 - P(\overline{A}_1)\,P(\overline{A}_2)\ldots P(\overline{A}_n)$

How is easy to prove using the above result

i.e., $$P(\overline{A}_1 \cap \overline{A}_2) = P(\overline{A}_1)\,P(\overline{A}_2)$$

3. Mutually independent events are always pair waise independent but converse is not necessarily true

CONDITIONAL PROBABILITY

If A & B are 2 events related to an experiment then the probability of occurence of A under the condition that B has already occured etc. i.e. relative to B, is known as conditional probability.

It is denoted by P(A/B) or P(A given B).

Note here $P(B) \neq 0$ is must.

What do we mean by this ?

Let us take an experiment to consider it. Two dices are thrown and 2 events A & B are recorded.

A : The sum of dices is 8.

B : There is an odd number on first die.

So now if we need to find the probability of occurence of event A such that B has already occured will be called conditional probability. "A given B".

i.e. P(A/B)

Now we can imagine from the above written lines that the sample space for P(A/B) will become the possible outcomes of event B.

And the favourable events become the intersection of A and B, as B has to be taken in consideration as result of A is based on B.

Hence the formula becomes

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

Let us complete the example we have taken

B = Occurence of odd on first die

= {(1, 1) (1, 2) (1, 3) .... (1, 6)
(3, 1) (3, 2) (3, 3) ... (3, 6)
(5, 1) (5, 2) (5, 3) ... (5, 6)}

= 18 cases

A = occurence of sum 8

= {(2, 6) (3, 5) (4, 4) (5, 3) (6, 2)}

= 5 cases

Note : Here (2, 6) and (6, 2) are 2 different cases and first represents 2 on dice 1 and 6 on dice 2. Next represents 6 on dice 1 and 2 on dice 2.

$$P(A) = \frac{5}{36} \text{ and } P(B) = \frac{18}{36} = \frac{1}{2}$$

{as sample space = 36, when 2 dices are rolled}

$n(A \cap B) = 2$ {possible cases are (3, 5) & (5, 3)

$$P(A \cap B) = \frac{2}{36} = \frac{1}{8}$$

Hence $$P(A/B) = \frac{n(A \cap B)}{n(B)} = \frac{2}{18} = \frac{1}{9}$$

$$\text{Or } = \frac{P(A \cap B)}{P(B)} = \frac{1}{18 \times \frac{1}{2}} = \frac{1}{9}$$

For independent events : If A and B are two independent events then

$$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)\,P(B)}{P(B)} = P(A)$$

as $$\mathbf{P(A \cap B) = P(A) \,.\, P(B)}$$

Similarly for $$P(B/A) = P(B)$$

MULTIPLICATION THEOREM

If A and B are two events associated with a random experiment, then

$$P(A \cap B) = P(A) \,.\, P(B/A)$$

or $$P(B \cap A) = P(B) \,.\, P(A/B)$$

This can be just derived from conditional probability.

Note : This theorem is used in cases where the events occur in different stages and not simultaneously.

EXTENSION OF THE THEOREM

$P(A_1 \cap A_2 \cap A_3) = P(A_1) \,.\, P(A_2/A_1) \,.\, P(A_3/(A_1 \cap A_2)$ and this can be generalized for n such events

Illustration 26

If A and B are two events such that P(A) = 0.5, P(B) = 0.6 and P(A ∪ B) = 0.8, find P(A/B) and P(B/A).

Solution :

We have, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$\therefore \quad P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.5 + 0.6 - 0.8 = 0.3$

Now, $$P(A/B) = \frac{P(A \cap B)}{P(B)} \Rightarrow P(A/B) = \frac{0.3}{0.6} = \frac{1}{2}$$

and $$P(B/A) = \frac{P(A \cap B)}{P(A)} \Rightarrow P(B/A) = \frac{0.3}{0.5} = \frac{3}{5}$$

Illustration 27

A die is rolled twice and the sum of the numbers appearing on them is observed to be 7. What is the conditional probability that the number 2 has appeared at least once ?

Solution :

Consider the following events :

A = getting number 2 at least once; B = getting 7 as the sum of the numbers on two dice. We have

A = {(2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), (1, 2), (3, 2), (4, 2), (5, 2), (6, 2)}

and B = {(2, 5), (5, 2), (6, 1), (1, 6), (3, 4), (4, 3)}

$\therefore \quad P(A) = \frac{11}{36}$, $P(B) = \frac{6}{36}$ and $P(A \cap B) = \frac{2}{36}$

Now, required probability

$$= P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{2/36}{6/36} = \frac{1}{3}$$

Illustration 28

Two dice are thrown. Find the probasbility that the sum of the numbers coming up on them is 9, if it is known that the number 5 always occurs on the first die.

Solution :

When two dice are thrown, let S be the sample space and

A = the event that the sumof the numbers coming up is 9.

and B = the event of occurrence of 5 on the first die.

Then S = {1,2, 3, 4, 5, 6} {1,2,3, 4, 5, 6}

$\therefore \quad n(S) = 6 \quad 6 = 36$

A = {(6, 3), (5, 4), (4, 5), (3, 6)} $\therefore$ n(A) = 4

and B = {(5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6)}

$\therefore \quad A \cap B = \{(5, 4)\} \therefore n(A \cap B) = 1$

Now $P(A \cap B) = \frac{n(A \cap B)}{n(S)} = \frac{1}{36}$ and $P(A) = \frac{n(A)}{n(S)} = \frac{4}{36}$

$\therefore$ Required probability, $P(A/B) = \frac{P(A \cap B)}{P(A)} = \frac{\frac{1}{36}}{\frac{4}{36}} = \frac{1}{4}$

Second Method : $P(A/B) = \frac{n(A \cap B)}{n(B)} = \frac{1}{4}$

Illustration 29

Two coins are tossed. What is the probability of coming up of two heads, if it is known that at least one head comes up ?

Solution :

When two coins are thrown, let S be the sample space and

A = the event of coming up of two heads

B = the event of coming upof at least one head

Then $S = \{(H, H), (H, T), (T, H), (T, T)\}$

$A = \{(H, H)\}$ and $B = \{(H, H), (H, T), (T, H)\}$

$\therefore$ $n(S) = 4,\ n(B) = 3,\ n(A) = 1$

Also $A \cap B = \{(H, H)\}$ $\therefore$ $n(A \cap B) = 1$

Now $P(B = \frac{n(B)}{n(S)} = \frac{3}{4}$ and $P(A \cap B) = \frac{n(A \cap B)}{n(S)} = \frac{1}{4}$

$\therefore$ Required probability, $P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3}$

Second Method : $P(A/B) = \frac{n(A \cap B)}{n(B)} = \frac{1}{3}$

Illustration 30

In a class 30% students fail in English; 20% students fail in Hindi and 10% students fail in English and Hindi both. A student is chosen at random, then what is the probability that he will fail in English if he has failed in Hindi ?

Solution :

Here random experiment is selection of a student.

Let S = the sample space

A = the event that the student chosen fails in English

and B = the event that the student chosen fails in Hindi.

Then $A \cap B$ = the event that the student chosen fails in both English and Hindi.

According to question, $n(S) = 100$, $n(B) = 20$ and $n(A \cap B) = 10$

$\therefore$ $P(B) = \frac{n(B)}{n(S)} = \frac{20}{100} = \frac{1}{5}$

and $P(A \cap B) = \frac{n(A \cap B)}{n(S)} = \frac{10}{100} = \frac{1}{10}$

$\therefore$ Required probability, $P(A/B) = \dfrac{P(A \cap B)}{P(B)} = \dfrac{\frac{1}{10}}{\frac{1}{5}} = \dfrac{1}{2}$

Second Method : $P(A/B) = \dfrac{n(A \cap B)}{n(B)} = \dfrac{10}{20} = \dfrac{1}{2}$

Illustration 31

A bag contains 5 white and 8 red balls. Two draws of 3 balls each are made without replacement. What is the probability that the first draw gives 3 white balls and second draw gives 3 red balls ?

Solution :

Let A = the event of drawing 3 white balls in the first draw and

B = the event of drawing 3 red balls in the second draw.

Then $A \cap B$ = the event of drawing 3 white balls in the first draw and 3 red balls in the second draw.

Now $P(A) = \dfrac{{}^5C_3}{{}^{13}C_3} = \dfrac{5}{143}$

when 3 white balls are drawn in the first draw, number of balls left = 10 and number of red balls left = 8

$\therefore$ $P(B/A) = \dfrac{{}^8C_3}{{}^{10}C_3} = \dfrac{7}{15}$

Now required probability,

$$P(A \cap B) = P(A) \,.\, P\left(\frac{B}{A}\right) = \frac{5}{143} \times \frac{7}{15} = \frac{7}{429}$$

Illustration 32

A bag X contains 3 white and 2 black balls; another bag Y contains 2 white and 4 black balls. A bag and a ball out of it is picket at random. What is the probability that the ball is white ? [I.T.T. 1971]

Solution :

Let E_1 = the event of selecting bag X

E_2 = the event of solecting bag Y

E = the event of drawing a white ball

Let $A = E_1 \cap E$ and $B = E_2 \cap E$

Now required probability,

$$P(E) = P(A \cup B) = P(A) + P(B)$$

[Since events A and B are mutually exclusive]

$$= P(E_1 \cap E) + P(E_2 \cap E)$$

$$= P(E_1) \,.\, P(E/E_1) + P(E_2) \,.\, P(E/.E_2)$$

$$= \frac{1}{2}.\frac{3}{5} + \frac{1}{2}.\frac{2}{6} = \frac{3}{10} + \frac{1}{6} = \frac{9+5}{30} = \frac{7}{15}$$

Illustration 33

A lot contains 50 defective and 50 non-defective bulbs. Two bulbs are drawn at random, one at a time, with replacement. The events A, B, C are defined as

A : "the first bulb is defective",

B : "the second bulb is non-defective",

C : "the two bulbs are bothdefective or both non-defective."

Determine whether

(i) **A, B, C are pairwise independent,**

(ii) **A, B, C are mutually independent.**

Solution :

We have $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{2}$, $P(C) = \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2} = \frac{1}{2}$.

$P(A \cap B)$ = Probability that the first is defective and the second is non-defective

$$= \frac{1}{2} \times \frac{1}{2} = \frac{1}{4} = P(A) \,.\, P(B)$$

$P(B \cap C)$ = Probability that both the bulbs are non-defective

$$= \frac{1}{2} \times \frac{1}{2} = \frac{1}{4} = P(B)\ P(C)$$

and $P(A \cap C)$ = Probability that both the bulbs are defective

$$= \frac{1}{2} \times \frac{1}{2} = \frac{1}{4} = P(A)\ P(C)$$

Hence, A, B, C are pairwise independent.

Now, $P(A \cap B \cap C)$ = Probability that the first bulb is defective and the second is non-defective and the first and second are both defective or both non-defective = 0

and $P(A) \,.\, P(B) \,.\, P(C) = \frac{1}{8}$

$\therefore$ $P(A \cap B \cap C) \neq P(A)\ P(B)\ P(C)$

Thus, A, B, C are not mutually independent.

Illustration 34

There are three urns A, B and C. Urn A contains 4 white balls and 5 blue balls. Urn B contains 4 white balls and 3 blue balls.Urn C contains 2 white balls and 4 blue balls. One ball is drawn from each of these urns. What is the probability that out of these three balls drawn, two are white balls and one is a blue ball ?

Solution :

Consider the following events :

E_1 = ball drawn from urn A is white,

E_2 = ball drawn from urn B is white,

E_3 = ball drawn from urn C is white

Then $P(E_1) = \frac{4}{9}$, $P(E_2) = \frac{4}{7}$ and $P(E_3) = \frac{2}{6} = \frac{1}{3}$

$\therefore$ $P(\overline{E}_1)$ = ball drawn from urn A is black

$= 1 - P(E_1) = 1 - \frac{4}{9} = \frac{5}{9}$,

$P(\overline{E}_2)$ = ball drawn from urn B is black

$= 1 - P(E_2) = 1 - \frac{4}{7} = \frac{3}{7}$

and $P(\overline{E}_3)$ = ball drawn from urn C is black

$= 1 - P(E_3) = 1 - \frac{1}{3} = \frac{2}{3}$

Now, two white balls and one black ball can be drawn in the followingmutually exclusive ways :

(I) White from urn A, white from urn B and black from urn C i.e. $E_1 \cap E_2 \cap \overline{E}_3$

(II) White from urn A, black from urn B and white from urn C i.e. $E_1 \cap \overline{E}_2 \cap E_3$

(III) Black from urn A, white from urn B and white from urn C i.e. $\overline{E}_1 \ldots E_2 \ldots E_3$

$\therefore$ Required probability = P(I) + P(II) + P(III)

$= P(E_1 \cap E_2 \cap \overline{E}_3) + P(E_1 \cap \overline{E}_2 \cap E_3) + P(\overline{E}_1 \cap E_2 \cap E_3)$

$= P(E_1)\, P(E_2)\, P(\overline{E}_3) + P(E_1\, P(\overline{E}_2)\, P(E_3) + P(\overline{E}_1)\, P(E_2)\, P(E_3)$

[. E_1, E_2, E_3 are independent events]

$= \frac{4}{9} \times \frac{4}{7} \times \frac{2}{3} + \frac{4}{9} \times \frac{3}{7} \times \frac{1}{3} + \frac{5}{9} \times \frac{4}{7} \times \frac{1}{3} = \frac{64}{189}$

Illustration 35

A and B throw alternately a pair of dice. A wins if he throws 6 before B throws 7 and B wins if he throws 7 before A throws 6. Find their respective chance of winning, if A begins.

Solution :

6 can be thrown with a pair of dice in the following ways : (1, 5), (5, 1), (4, 2), (2, 4), (3, 4).

So, probability of throwing a '6' $= \frac{5}{36}$

and, probability of not throwing a '6' $= 1 - \frac{5}{36} = \frac{31}{36}$

Now, 7 can be thrown with a pair of dice in 6 ways, viz. (1, 6), (6, 1), (2, 5), (5, 2), (4, 3), (3, 4).

So, probability of throwing a '7' $= \frac{6}{36} = \frac{1}{6}$

and, probability of not throwing a '7' $= 1 - \frac{1}{6} = \frac{5}{6}$.

Let E and F be two events defined as :

E = throwing a '6' in a single throw of a pair of dice, and

F = throwing a '7' in a single throw of a pair of dice.

Then $P(E) = \frac{5}{36}$, $P(\overline{E}) = \frac{31}{36}$, $P(F) = \frac{1}{6}$ and $P(\overline{E}) = \frac{5}{6}$

A wins if he throws '6' in 1st or 3rd or 5th ..... throws.

Probability of A throwing a '6' in first throw $= P(E) = \frac{5}{36}$.

A will get third if he fails in first and B fails in second throw.

$\therefore$ Probability of A throwing a '6' in third throw

$$P(\overline{E} \cap \overline{F} \cap E) = P(\overline{E})\, P(\overline{F})\, P(E) = \frac{31}{36} \times \frac{5}{6} \times \frac{5}{36}$$

Similarly, probability of A throwing a '6' in fifth throw

$$= P(\overline{E} \cap \overline{F} \cap \overline{E} \cap \overline{F} \cap E) = P(\overline{E})\, P(\overline{F})\, P(\overline{E})\, P(\overline{F})\, P(E)$$

$$= \left(\frac{31}{36}\right)^2 \times \left(\frac{5}{6}\right)^2 \times \frac{5}{36} \text{ and so on}$$

Hence, probability of winning of A

$$= P(E \cup (\overline{E} \cap \overline{F} \cap E) \cup (\overline{E} \cap \overline{F} \cap \overline{E} \cap \overline{F} \cap E) \cup \ldots]$$

$$= P(E) + P(\overline{E} \cap \overline{F} \cap E) + P(\overline{E} \cap \overline{F} \cap \overline{E} \cap \overline{F} \cap E) + \ldots$$

$$= \frac{5}{36} + \left(\frac{31}{36} \times \frac{5}{6}\right) \times \frac{5}{36} + \left(\frac{31}{36} \times \frac{5}{6}\right)^2 \times \frac{5}{36} + \ldots$$

$$= \frac{5/36}{1 - (31/36) \times (5/6)} = \frac{30}{61}$$

Thus, probability of winning of B = $1 - \frac{30}{61} = \frac{31}{61}$

THE THEOREM OF TOTAL PROBABILITY

To establish this result we start with the definition of a partition of a sample space.

A partition of a sample space

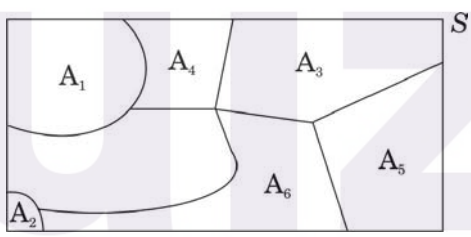
The collection of events $A_1, A_2, \ldots A_n$ is said to **partition** a sample space S if

(a) $A_1 \cup A_2 \cup \ldots \cup A_n = S$

(b) $A_i \cap A_j = \phi$ for all i, j

(c) $A_i \neq \phi$ for all i

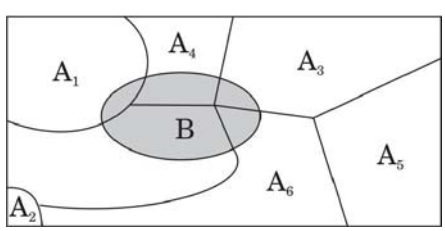
In essence, a partition is a collection of non-empty, non-overlapping subsets of a sample space whose union is the sample space itself.



If B is any event within S then we can express B as the unionof subsets :

$$B = (B \cap A_1) \cup (B \cap A_2) \cup \ldots \cup (B \cap A_n)$$

The definition is illustrated in Figure in which an event B in S is represented by the shaded region.



The bracketed events $(B \cap A_1)$, $(B \cap A_2)$... $(B \cap A_n)$ are mutually exclusive (if one occurs then none of the others can occur) and so, using the addition law of probability for mutually exclusive events :

$$P(B) = P(B \cap A_1) + P(B \cap A_2) + \ldots + P(B \cap A_n)$$

Each of the probabilities on the right-hand side may be expressed in terms of conditional probabilities :

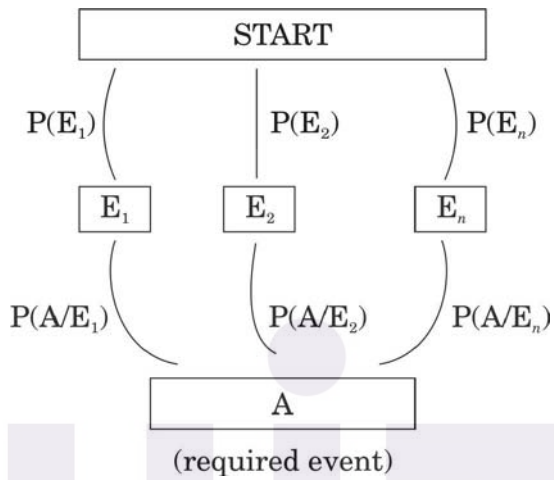
$$P(B \cap A_i) = P(B/A_i)\ P(A_i)$$

$$P(B) = P(B \mid A_1)\ P(A_1) + P(B \mid A_2)\ P(A_2) + \ldots + P(B \mid A_n)\ P(A_n)$$

$$= \sum_{i=1}^{n} P(B \mid A_i) P(A_i)$$

This is the theorem of Total Probability. A related theorem with many applications in statistics can be deduced from this, known as Bayes' theorem.

We solve the questions on total probability by tree diagram.



So total probability of event A is written as

$$P(A) = P(E_1)\ P(A \mid E_1) + P(E_2)\ P(A \mid E_2) \ldots + P(E_n)\ P(A \mid E_n)$$

Illustration 36

A box contains N coins, m of which are fair and the rest are biased. The probability of getting a head when a fair coin is tossed is 1/2, while it is 2/3 when a biased coin is tossed. A coin is drawn from the box at random and is tossed twice. The first time it shows head and the second time it shows tail. What is the probability that the coin drawn is fair ? [I.I.T. 2002]

Solution :

Let E_1 be the event that the coin drawn is fair and E_2 be the event that the coin drawn is biased.

$$\therefore \qquad P(E_1) = \frac{m}{N} \text{ and } P(E_2) = \frac{N-m}{N}$$

A is the event that on tossing the coin the head appears first and then appears tail.

$$\therefore \qquad P(A) = P(E_1 \cap A) + P(E_2 \cap A)$$

$$= P(E_1)\ P\ (A/E_1) + P(E_2)\ P(A/E_2)$$

$$= \frac{m}{N}\left(\frac{1}{2}\right)^2 + \left(\frac{N-m}{N}\right)\left(\frac{2}{3}\right)\left(\frac{1}{3}\right) \qquad \ldots(1)$$

We have to find the probability that A has happened because of E_1

$$\therefore \qquad P(E_1/A) = \frac{P(E_1 \cap A)}{P(A)}$$

$$= \frac{\frac{m}{N}\left(\frac{1}{2}\right)^2}{\frac{m}{N}\left(\frac{1}{2}\right)^2 + \frac{N-m}{N}\left(\frac{2}{3}\right)\left(\frac{1}{3}\right)} \qquad \text{(by (1))}$$

$$= \frac{m/4}{m/4 + \frac{2(N-m)}{9}} = \frac{9m}{m+8N}$$

Illustration 37

For a student to qualify, he must pass at least two out of three exams. The probability that he will pass the 1st exam is p. If he fails in one of the exams then the probability of his passing in the next exam is $\frac{p}{2}$ otherwise it remains the same. Find the probability that he will qualify. [I.I.T. 2003]

Solution :

Let us consider

$E_1 \equiv$ event of passing I exam.
$E_2 \equiv$ event of passing II exam.
$E_3 \equiv$ event of passing III exam.

Then a student can qualify in anyone of following ways

1. He passes first and second exam.
2. He passes first, fails in second but passes third exam.
3. He fails in first, passes second and third exam.

$\therefore$ Required probability

$= P(E_1) + P(E_2/E_1) + P(E_1)\, P(E_2/E_1)\, P(E_3/E_2) + P(E_1)\, P(E_2/E_1)\, P(E_3/E_2)$

[as an event is dependent on previous one]

$$= p.p + p.\,(1-p)\,.\frac{p}{2} + (1-p)\,.\frac{p}{2}.\,p$$

$$= p^2 + \frac{p^2}{2} - \frac{p^3}{2} + \frac{p^2}{2} - \frac{p^3}{2} = 2p^2 - p^3$$

Illustration 38

An urn contains 2 white and 2 black balls. A ball is drawn at random. If it is white it is not replaced into the urn. Otherwise it is replaced along with other balls of same colour. The process is repeated. Find the probability that 3rd ball drawn is black. [IIT 1987]

Solution :

(a) There are following four possible ways of drawing first two balls

(i) Both the first and the second balls drawn are white.

(ii) The first ball drawn is white and the second ball drawn is black.

(iii) The first ball drawn is black and the second ball drawn is white.

(iv) Both the first and the second balls drawn are black. Let us define events (i), (ii), (iii) and (iv) by E_1, E_2, E_3, E_4 respectively. Also let E denote the event that the third ball drawn is black.

Then, $P(E_1) = \frac{2}{4} \times \frac{1}{3} = \frac{1}{6}$, $P(E_2) = \frac{2}{4} \times \frac{2}{3} = \frac{1}{3}$

$$P(E_3) = \frac{2}{4} \times \frac{2}{5} = \frac{1}{5}$$

$$P(E_4) = \frac{2}{4} \times \frac{3}{5} = \frac{3}{10}$$

Also $P(E/E_1) = 1$, since when the event E, has alreaqdy happened i.e., the first two balls drawn are both white, they are not replaced and so there are left 2 block balls in the urn so that the probability that the third ball drawn in this case is black $= \frac{2}{2} = 1$.

Again $P(E/E_2) = \frac{3}{4}$, since when the event E_2 has already happened there are 3 black and one white balls in the urn. So in this case the probability that the third ball drawn is black $= \frac{3}{4}$.

Similarly, $P(E/E_3) = \frac{3}{4}$ and $P(E/E_4) = \frac{2}{3}$

Now by theorem of total prob. for compound events, we have

$$P(E) = P(E_1)\,P(E/E_1) + P(E_2)\,P(E/E_2) + P(E_3)\,P(E/E_3) + P(E_4)\,P(E/E_4)$$

$$= \frac{1}{6} \times 1 + \frac{1}{3} \times \frac{3}{4} + \frac{1}{5} \times \frac{3}{4} + \frac{3}{10} \times \frac{2}{3}$$

$$= \frac{1}{6} + \frac{1}{4} + \frac{3}{20} + \frac{1}{5} = \frac{23}{30}$$

Illustration 39

A lot contains 20 articles. The probability that the lot contains exactly 2 defective articles is 0.4 and the probability that a lot contains exactly 3 defective articles is 0.6. Articles are drawn from the lot at random one by one without replacement and are tested till defective articles are found. What is the probability that testing procedure ends at the twelfth testing. [IIT 1986]

Solution :

Let A_1 be the event that the lot contains 2 defective articles and A_2 the event that the lot contains 3 defective articles. Also let A be the event that the testing procedure ends at the twelfth testing. Then according to the question :

$$P(A_1) = 0.4 \text{ and } P(A_2) = 0.6$$

Since $0 < P(A_1) < 1$, $0 < P(A_2) < 1$, and $P(A_1) + P(A_2) = 1$

$\therefore$ The events A_1, A_2 form a partition of the sample space. Hence by the theorem of total probability for compound events, we have

$$P(A) = P(A_1)\, P(A/A_1) + P(A_2)\, P(A/A_2) \qquad ...(1)$$

Hence $P(A/A_1)$ is the probability of the event the testing procedure ends at the twelfth testing when the lot contains 2 defective articles. This is possible when out of 20 articles, first 11 draws must contain 10 non defective and 1 defective article and 12th draw must give a defective article.

$$\therefore \qquad P(A/A_1) = \frac{{}^{18}c_{10} \times {}^{2}c_1}{{}^{20}c_{11}} \times \frac{1}{9} = \frac{11}{190}$$

Similarly, $$P(A/A_2) = \frac{{}^{17}c_9 \times {}^{3}c_1}{{}^{20}c_{11}} \times \frac{1}{9} = \frac{11}{228}$$

Now substituting the values of $P(A/A_1)$ and $P(A/A_2)$ in eq. (1) we get

$$p(A) = 0.4 \quad \frac{11}{190} + 0.6 \times \frac{11}{228} = \frac{11}{475} + \frac{11}{380} = \frac{99}{1900}$$

Illustration 40

In a certain city only 2 newspapers A & B are published, it is known that 25% of the city population reads A and 20% reads B while 8% reads both A & B. It is also known that 30% of those who read A but not B look into advertisements and 40% of those who read B but not A look into advertisements, while 50% of those who read both A and B look into advertisements. What is the percentage of the population that reads advertisement ?

[IIT 1984]

Solution :

Let P(A) denotes the prob. of people reading newspaper A and P(B) that of people reading newspaper B.

Then, $$P(A) = \frac{25}{100} = 0.25$$

$$P(B) = \frac{20}{100} = .20,\ P(AB) = \frac{8}{100} = 0.08$$

Prob. of people reading the newspaper A but not

$$B = P(AB^c) = P(A) - P(AB)$$
$$= 0.25 - 0.08 = 0.17$$

Similarly,

$$P(A^cB) = P(B) - P(AB) = 0.20 - 0.08 = 0.12$$

Let E be the event that a person reads an advertisement. Therefore

according to question $P(E/AB^c) = \frac{30}{100}$ $P(E/A^cB) = \frac{40}{100}$

$$P(E/AB) = \frac{50}{100}$$

$\therefore$ By total prob. theorem

$$P(E) = P(E/AB^c)\ P(AB^c) + P(E/A^cB)\ P(A^cB) + P(E/AB)\ .\ P(AB)$$
$$= \frac{30}{100} \times 0.17 + \frac{40}{100} \times 0.12 + \frac{50}{100} \times 0.08$$
$$= 0.051 + 0.048 + 0.04$$
$$= 0.139.$$

Thus the population that reads advertisement is 13.9%. **Ans.**

BAYES' THEOREM

We again consider the conditional probability statement :

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A \cap B)}{P(B \mid A_1)\ P(A_1) + P(B \mid A_2)\ P(A_2) + \ldots + P(B \mid A_n)\ P(A_n)}$$

in which we have used the theorem of Total Probability to replace P(B). Now

$$P(A \cap B) = P(B \cap A) = P(B \mid A) \quad P(A)$$

Substituting this in the expression for $P(A \mid B)$ we immediately obtain the result

$$P(A \mid B) = \frac{P(B \mid A) \times P(A)}{P(B \mid A_1)P(A_1) + P(B \mid A_2)P(A_2) + \ldots + P(B \mid A_n)P(A_n)}$$

This is true for any event A and so, replacing A by A_i gives the result, known as Bayes' theorem as

$$P(A_i \mid B) = \frac{P(B \mid A_i) \times P(A_i)}{P(B \mid A_1)\ P(A_1) + P(B \mid A_2)\ P(A_2) + \ldots + P(B \mid A_n)\ P(A_n)}$$

This theorem is used in the cases where the required even thas already taken place and you are required to find the probability under some previous stage conditions.

For example.

The required event A has occured then find the probability that it occured through event B (which is defined in prior stages)

Illustration 41

In a test an examine either guesses or copies of knows the answer to a multiple choice question with four choices. The probability that he make a guess is 1/3 and the probability that he copies the answer is 1/6. The probability that his answer is correct given that he copied it, is 1/8. Find the probability that he knew the answer to the question given that he correctly answered it. [I.I.T. 1991]

Solution :

First of all to find out whether this question is of Baye's theorem or not.

See the language of the question, "he correctly answered it" i.e. the event has already occured. And we asked to find the probability of occured event on some prior event i.e. "he knows the answer".

Let us define different events so as to ease out the question.

$A_1 \equiv$ the examinee guesses the answer

$A_2 \equiv$ the examine copies the answer

$A_3 \equiv$ the examinee knows the answer

$A \equiv$ the examinee answers correctly

Now according to question

$$P(A_1) = \frac{1}{3}$$

$$P(A_2) = \frac{1}{6}$$

Now notice that A_1 A_2 A_3 are mutually exclusive and exhaustive event.

i.e. $P(A_1) + P(A_2) + P(A_3) = 1$

we needed this to find the value of $P(A_3)$

$$\Rightarrow \quad \frac{1}{3} + \frac{1}{6} + P(A_3) = 1$$

$$\Rightarrow \quad P(A_3) = \frac{1}{2}$$

We need to find the probability of $P(A_3/A)$ which is equal to

$$P(A_3/A) = \frac{P(A/A_3)\,P(A_3)}{P(A/A_1)\,P(A_1) + P(A/A_2)\,P(A_2) + P(A/A_3)\,P(A_3)} \quad ...(i)$$

for this we need the values of $P(A/A_1)$ $P(A/A_2)$ $P(A/A_3)$

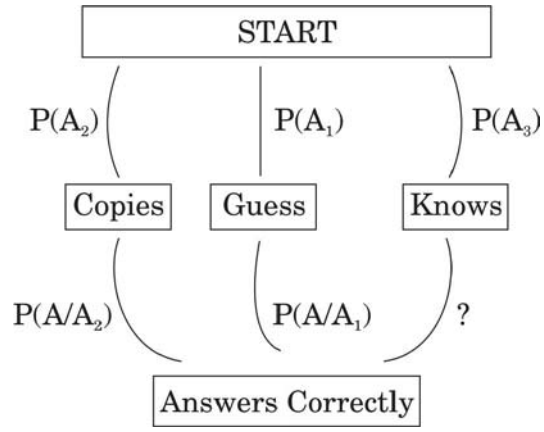
seeing this from tree diagram we are given with $P(A/A_2)$ i.e. he answers correctly after copying

$$P(A/A_2) = \frac{1}{8}$$

Now coming to $P(A/A_1)$

it is equal to $\frac{1}{4}$

as there are only 4 options to guess from and one option will be correct.



for $P(A/A_3)$, it is equal to 1, obviously because he knows the correct answer which can be known in 1 possible way only.

Now substituting all these values in (i)

$$P(A_3/A) = \frac{P(A/A_3)\,P(A_3)}{P(A/A_1)\,P(A_1) + P(A/A_2)\,P(A_2) + P(A/A_3)\,P(A_3)}$$

$$= \frac{1.\frac{1}{2}}{\frac{1}{4}.\frac{1}{3} + \frac{1}{8}.\frac{1}{6} + 1.\frac{1}{2}}$$

$$= \frac{1}{2} \times \frac{48}{29} = \frac{24}{29}$$

Illustration 42

Three urns contain 6 red, 4 black, 4 red, 6 black and 5 red and 5 black balls respectively. One of the urns is selected at random and a ball is drawn from it. If the ball drawn is red, find the probability that is drawn from the first urn.

Solution :

Let E_1, E_2, E_3 and A be the events defined as follows:

E_1 = urn first is chosen, E_2 = urn second is chosen,

E_3 = urn third is chosen, and A = ball drawn is red.

Since there are three urns and one of the three urns is chosen at random, therefore

$$P(E_1) = P(E_2) = P(E_3) = \frac{1}{3}.$$

If E_1 has already occurred, then urn first has been chosen which contains 6 red and 4 black balls. The probability of drawing a red ball from it is 6/10.

So, $P(A/E_1) = \frac{6}{10}$. Similarly

$$P(A/E_2) = \frac{4}{10} \text{ and } P(A/E_3) = \frac{5}{10}$$

We are required to find $P(E_1/A)$, i.e. given that the ball drawn is red, what is the probability that it is drawn from the first urn.

By Baye]s rule

$$P(E_1/A) = \frac{P(E_1)\,P(A/E_1)}{P(E_1)\,P(A/E_1) + P(E_2)\,P(A/E_2) + P(E_3)\,P(A/E_3)}$$

$$= \frac{\frac{1}{3} \times \frac{6}{10}}{\frac{1}{3} \times \frac{6}{10} + \frac{1}{3} \times \frac{4}{10} + \frac{1}{3} \times \frac{5}{10}} = \frac{6}{15} = \frac{2}{5}$$

Illustration 43

A man is known to speak truth 3 out of 4 times. He throws a die and reports that it is a six. Find the probability that its actually a six.

Solution :

Let E_1, E_2 and A be the events defined as follows :

E_1 = six occurs, E_2 = six does not occur, and A = the man reports that it is a six.

We have $P(E_1) = \frac{1}{6}$, $P(E_2) = \frac{5}{6}$

Now, $P(A/E_1)$ = Probability that the man reports that the is a six on the die given that six has occurred on the die

= Probability the man speaks truth = 3/4 and $P(A/E_2)$

= Probability that the man reports that there is on the die given that six has not occurred on the die.

= Probability that the man does not speak truth

$$= 1 - \frac{3}{4} = \frac{1}{4}$$

We have to find $P(E_1/A)$ i.e., the probability that there is six on the die given that the man has reported that there is six. By Baye's rule, we have

$$P(E_1/A) = \frac{P(E_1)\,P(A/E_1)}{P(E_1)\,P(A/E_1) + P(E_2)\,P(A/E_2)} = \frac{\frac{1}{6} \times \frac{3}{4}}{\frac{1}{6} \times \frac{3}{4} + \frac{5}{6} \times \frac{1}{4}} = \frac{3}{8}$$

Illustration 44

A person goes to office either by car, scooter, bus or train, the probability of which being $\frac{1}{7}, \frac{3}{7}, \frac{2}{7}$ and $\frac{1}{7}$ respectively. Probability that he reaches office late, if he takes car, scooter, bus or train is $\frac{2}{9}, \frac{1}{9}, \frac{4}{9}$ and $\frac{1}{9}$ respectively. Given that he reached office in time, then what is the probability that he travelled by a car.

Solution :

Let us define the following events

C ≡ person goes by car,

S ≡ person goes by scooter

B ≡ person goes by bus,

T ≡ person goes by train,

L ≡ person reaches late

Then we aqre given in the question

$$P(C) = \frac{1}{7};\ P(S) = \frac{3}{7};\ P(B) = \frac{2}{7};\ P(T) = \frac{1}{7}$$

$$P(L/C) = \frac{2}{9};\ P(L/S) = \frac{1}{9};\ P(L/B) = \frac{4}{9};\ P(L/T) = \frac{1}{9}$$

To find the prob. $P(C/\bar{L})$ [. reaches in time ≡ not late]

Using Baye's theorem

$$P(C/\bar{L}) = \frac{P(\bar{L}/C)\,P(C)}{P(\bar{L}/C)\,P(C) + P(\bar{L}/S)\,P(S) + P(\bar{L}/B)\,P(B) + A(\bar{L}/T)\,P(T)} \quad ...(i)$$

Now, $P(\bar{L}/C) = 1 - \frac{2}{9} = \frac{7}{9}$; $P(\bar{L}/S) = 1 - \frac{1}{9} = \frac{8}{9}$

$$P(\bar{L}/B) = 1 - \frac{4}{9} = \frac{5}{9} \qquad P(\bar{L}/T) = 1 - \frac{1}{9} = \frac{8}{9}$$

Substituting these values, we get

$$P(C/\bar{L}) = \frac{\frac{7}{9} \times \frac{1}{7}}{\frac{7}{9} \times \frac{1}{7} + \frac{8}{9} \times \frac{3}{7} + \frac{5}{9} \times \frac{2}{7} + \frac{8}{9} \times \frac{1}{7}}$$

$$= \frac{7}{49} = \frac{1}{7}$$

Illustration 45

A letter is known to have come either from TATANAGAR or CALCUTTA. On the envelope just two consecutive letters TA are visible. What is the probability that the letter has come from (i) Calcutta (ii) Tatanagar ?

Solution :

Let E_1 be the event that the letter came from Calcutta an E_2 be the event that the letter came from Tatanagar. Let A denote the event that two consecutive letters visible on the envelope are TA.

Since the letters have comeeither from Calcutta or Taganagar, therefore,

$$P(E_1) = \frac{1}{2} = P(E_2).$$

If E_1 has occurred, then it means that the letter came from Calcutta. In the word CALCUTTA there are 8 letters in which TA occurs in the end. Considering TA as one letter there are seven letters out of which one can be in 7 ways. Therefore

$$P(A/E_1) = \frac{1}{7}$$

If E_2 has occurred, then the letter came from Tatanagar. In the word TATANAGAR there are 9 letters in which TA occurs twice. Considering one of the two TA's as one letter there are 8 letters. Therefore

$$P(A/E_2) = \frac{2}{8}$$

By Baye's rule

(i) $$P(E_1/A) = \frac{P(E_1)\,P(A/E_1)}{P(E_1)\,P(A/E_1) + P(E_2)\,P(A/E_2)}$$

$$= \frac{\frac{1}{2} \times \frac{1}{7}}{\frac{1}{2} \times \frac{1}{7} + \frac{1}{2} \times \frac{2}{8}} = \frac{4}{11}$$

(ii) $$P(E_2/A) = \frac{P(E_2)\,P(A/E_2)}{P(E_1)\,P(A/E_1) + P(E_2)\,P(A/E_2)}$$

$$= \frac{\frac{1}{2} \times \frac{2}{8}}{\frac{1}{2} \times \frac{1}{7} + \frac{1}{2} \times \frac{2}{8}} = \frac{7}{11}$$

BINOMIAL DISTRIBUTION

In n independent trials of a random experiment, let X be the number of times an event A occurs. In each trial, event A has same probability as P(A) = p referred to as success. Then in a trial non-occurrence of A is referred as failure and given by q = 1 – p.

Here X can assumed values from 0 to n. Now X = r means A occurs in r trials and (n – r) it does not occur this may look as

$$\underbrace{A\ A\ A.....A}_{r\ \text{times}}\ \underbrace{\overline{A}\ \overline{A}.....\overline{A}}_{n-r\ \text{times}}$$

here $\overline{A}$ means complement of A. Using the assumption that trials are independent, that is, they do not influence each other, hence has the probability

$$\underbrace{p\ \ p.....p}_{r\ \text{times}}\ \underbrace{q\ \ q.....q}_{n-r\ \text{times}} = p^r\ \ q^{n-r}$$

and it can be arranged in $\dfrac{n\,!}{r\,!\,(n-r)\,!} = {}^nC_r$ ways.

Now, if X denotes the number of successes in 4 trials, there X can take values 0, 1, 2, 3 and 4 such that

$$P(X = 0) = {}^4C_0\ p^0\ q^{4-0},\ P(X = 1) = {}^4C_1\ p^1\ q^4 - 1,$$

$$P(X = 2) = {}^4C_2\ p^2\ q^{4-2},\ P(X = 3) = {}^4C_3\ p^3 q^{4-3}$$

and $$P(X = 4) = {}^4C_4\ p^4\ q^{4-4}.$$

In other words, $P(X = r) = {}^4C_r\ p^r\ q^{4-r}$; r = 0, 1, 2, 3, 4.

This result can be generalised to the case where the experiment is repeated n times under identical conditions. The probability of r successes in n independent trials in a specific order say SSSFSFFSS ... SF is given by r-successes, (n – r) failures

$$P(SSSFSFFSS\ ...\ SF) = P(S)\ P(S)\ P(S)\ P(F)\ ...\ P(S)\ P(F)$$

(by multiplication theorem)

$$= pppq\ ...\ p\ .\ q = p^r\ q^{n-r}$$

Hence the probability of getting r successes or occurrence of A in r trials out of n independent trials is

$$P(X = r) = {}^nC_r\ p^r\ q^{n-r}$$

which is denoted as binomial distribution of random variable X.

- The probability of getting atleast k successes in

$$P(X \geq k) = \sum_{r=k}^{n} {}^nC_r\ p^r\ q^{n-r}$$

- The probability of getting almost K successes is

$$P(X \le k) = \sum_{r=0}^{k} {}^nC_r \, p^r \, q^{n-r}$$

- $\sum_{r=0}^{n} {}^nC_r \, p^r \, q^{n-r} = (p + q)^n = 1$

Illustration 46

Ten coins are tossed simultaneously. Find the probability of getting at least 7 heads.

Solution :

In this case n = 10 ; p = 1/2; q = 1/2

$$P(X \ge 7) = P(X = 7) + P(X = 8) + P(X = 9) + P(X = 10)$$

$$= ({}^{10}C_3 + {}^{10}C_2 + {}^{10}C_1 + 1) \frac{1}{2^{10}} = \frac{176}{2^{10}}$$

Illustration 47

Numbers are chosen at random, one at a time, from the two digit numbers 00, 01, 02, ... 99 with replacement. An event E occurs if and only if the product of the two digits of a selected number is 18. If four numbers are selected, find the probability that E occurs at least 3 times.

Solution :

The numbers, whose two digits product is 18, are 29, 36, 63, 92

$$P(E) = \frac{4}{100} = 0.04$$

Since there is replacement, P(E) remains the same for every selection. Four such selections are made.

Probability that E occurs, at least thrice

= Probability that E occurs thrice and fails to occur once + Probability that E occurs all the four times

$$= {}^4C_3 \, (0.04)^3 \, (0.96) + (0.04)^4 = 0.00024832$$

Illustration 48

A man takes a step forward with probability 0.4 and backwards with probability 0.6. Find the probability that at the end of eleven steps he is just one step away from the starting point. (I.I.T. 1987)

Solution :

Let p denote the probability that the man takes a step forward. Then p = 0.4

$\therefore \qquad q = 1 - p = 1 - 0.4 = 0.6$

Let X denote the number of steps taken in the forward direction. Since the steps are independent of each other, therefore X is a binomial variate with parameters n = 11 and p = 0.4 such that

$$P(X = r) = {}^{11}C_r\ (0.4)^r\ (0.6)^{11-r};\ r = 0, 1, 2, ..., 11 \qquad ...(i)$$

Since the man is one step away from the initial point, he is either one step forward or one step backward from the initial point at the end of eleven steps. If he is one step forward, then he must have taken six steps forward and five steps backward and if he is one step backward, then he must have taken five steps forward and six steps backward. Thus, either X = 6 or X = 5.

$\therefore$ Required probability $= P[(X = 5) \text{ or } (X = 6)]$

$= P(X = 5) + P(X = 6)$

$= {}^{11}C_5\ (0.4)^5\ (0.6)^{11-5} + {}^{11}C_6\ (0.4)^6\ (0.6)^{11-6}$ [Using (i)]

$= {}^{11}C_5\ (0.4)^5\ (0.6)^5\ [0.6 + 0.4]$ $\qquad [\because\ {}^{11}C_5 = {}^{11}C_6]$

$= 462\ (0.4)^5\ (0.6)^5 = 462\ (0.24)^5$

Illustration 49

Suppose the probability for A to win a game against B is 0.4. If A has an option of playing a "best of 3 games" or a "best of 5 games" match against B, which option should A choose so that the probability of his winning the match is higher ? (No game ends in a draw).

[I.I.T. 1989]

Solution :

Let E = the event that A wins a game against B.

Let occurrence of the event E be called a success and X denote the number successes.

Lt E_1 = the event that A wins "a best of 3 games" match

A will win a best of 3 games match if he wins in 2 or 3 games.

E_2 = the event that A wins "a best of 5 games" match.

A will win a best of 5 games match if he wins in 3 or 4 or 5 games

Now, $P(E_1) = P(X = 2 \text{ or } X = 3)$

$= P(X = 2) + P(X = 3)$

$= {}^3C_2\ p^2q + {}^3C_3p^3$

$= 3(0.4)^2\ (0.6) + (0.4)^3$

$= (0.4)^2\ [0.18 + 0.4] = (0.4)^2\ (2.2) = 0.352.$

$P(E_2) = P(X = 3 \text{ or } X = 4 \text{ or } X = 5)$

$= P(X = 3) + P(X = 4) + P(X = 5)$

$= {}^5C_3p^3q^2 + {}^5C_4p^4q + {}^5C_5p^5$

$= (10p^3q^2 + 5p^4q + p^5 = p^3\ (10q^2 + 5pq + p^2)$

$= (0.4)^3 \quad [10 \quad (0.6)^2 + 5 \quad (0.4) \quad (0.6) + (0.4)^2]$

$= (0.064) \quad (3.6 + 1.2 + 0.16) = 0.064 \quad 4.96 = 0.317$

Since $P(E_1) > P(E_2)$, hence A should choose the first option.

Illustration 49

An urn contains 25 balls of which 10 balls bear a mark 'A' and the remaining 15 balls bear a mark 'B'. A ball is drawn at random from the urn, its mark is noted down and it is replaced. If 6 balls are drawn in this way, find the probability that

(i) **all will bear 'A' mark**

(ii) **not more than 2 will bear 'B' mark**

(iii) **the number of balls with 'A' mark and 'B' mark will be equal**

(iv) **at least one ball will bear 'B' mark**

Solution :

Let p denote the probability of drawing a ball which bears mark 'A'. Then

$$p = \frac{10}{25} = \frac{2}{5}$$

Let X denote the number of balls which bear mark 'A' in 6 draws. Then X is a binomial variate with parameters n = 6 and $p = \frac{2}{5}$. Also $q = 1 - p = 1 - \frac{2}{5} = \frac{3}{5}$

Now, P(X = r) = Probability of getting r balls bearing mark 'A'.

$$= {}^6C_r \left(\frac{2}{5}\right)^r \left(\frac{3}{6}\right)^{6-r}, \ r = 0, 1, 2, ..., 6 \qquad ...(i)$$

(i) Probability that all balls bear 'A' = P(X = 6)

$$= {}^6C_6 \left(\frac{2}{5}\right)^6 \left(\frac{3}{6}\right)^{6-6} \qquad \text{[Using (i)]}$$

$$= (2/5)^6.$$

(ii) Not more than 2 balls will bear 'B' mark means that there can be either no ball or one ball or two balls of 'B' mark. This implies that there can be either 6 or 5 or 4 balls of 'A' mark.

$\therefore$ Required probability = $P(X \geq 4)$

$$= P(X = 4) + P(X = 5) + P(X = 6)$$

$$= {}^6C_4 \left(\frac{2}{4}\right)^4 \left(\frac{3}{5}\right)^2 + {}^6C_5 \left(\frac{2}{5}\right)^5 \left(\frac{3}{5}\right) + {}^6C_6 \left(\frac{2}{5}\right)^6 \left(\frac{3}{5}\right)^0 = 7\left(\frac{2}{5}\right)^4$$

<u>ALITER</u> Let p denote the probability that a ball drawn bears mark 'B'. Then

$$p = \frac{15}{25} = \frac{3}{5}$$

Let Y denote the number of balls which bear mark 'B' in 65 draws. Then Y is a binomial variate with parameters n = 6 and p = 3/5 such that